Ranking AML Alerts Under a Fixed Review Budget

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Problem Statement

Banks spend billions on anti-money laundering (AML) compliance every year, and most of that money gets spent on labor. A LexisNexis study estimated the cost of financial crime compliance in the US and Canada to be about $61 billion, with labor as the largest share. Most of that labor time is taken up by reviewing alerts, and more often than not the alerts in the end turn out to be normal activity.

The core problem that I'm exploring is that an AML team can only review so many cases a day. For this project, I'm using a fictional mid-sized bank named Kestrel Bank, whose investigators can review 50 cases per day. Whatever system decides which 50 cases get put in front of the investigator to be reviewed ultimately determines how much laundering the bank actually finds and a poorly ranked system wastes a lot of the analysts hours on normal accounts, while laundering that never gets flagged is never investigated.

The cost of getting this wrong is real. In 2024, TD Bank pleaded guilty to Bank Secrecy Act violations and agreed to pay about $3 billion, after leaving 92% of its transaction volume completely unmonitored. My project won't claim that my model would have prevented that, since TD's failures were about coverage and governance. But this example does show that how a bank monitors transactions is not just a technical choice but actually a major business risk.

So my question is: with a fixed review budget, which monitoring upgrades actually catch more laundering, and are they worth what they cost? In my tests I will compare a rules-based system against 3 different machine learning models, then test whether network-based features or a feedback loop that learns from investigations help to improve detection over time.

Target Audience

My target audience will be Kestrel Bank's BSA/AML Officer and compliance leadership. They are the ones who decide where to spend the compliance budget, and they are ultimately the ones who have to answer to examiners if monitoring falls short. The decision I will be asking my audience to make is whether Kestrel should keep its simple rules-based monitoring, replace it with a machine learning model, or invest in a system that keeps learning from its own investigations. Model governance is another factor here too SR 26-2, the federal banking regulators April 2026 model risk guidance expects new models and model changes, including retraining, to be validated and overseen. This means that more complexity means more of that work, so I will try to prove that the upgrades I will potentially be suggesting will pay for its own oversight.

Dataset Selection and Data Dictionary

The dataset I'm using is SAML-D, a synthetic AML transaction dataset published by Oztas et al. (2023) and is available on Kaggle. I chose this dataset because it includes timestamps, payment types, bank locations, currencies, and a label for each laundering scheme, which lets me test detection over time and by different types of laundering. Another reason I chose this dataset is because it is packed with data, it contains 9,504,852 unique transactions over a 10.5 month span with a date range from Oct 7, 2022 - Aug 23, 2023. There are also 855,460 different individual accounts and 9,873 laundering transactions over 17 different laundering schemes which is only 0.10% of the dataset which is the kind of extreme imbalance real AML monitoring deals with.

Variable(s)	**Meaning**
Date; Time	Transaction date and time
Sender_account; Receiver_account	Sending and receiving account identifiers
Amount	Transaction amount
Payment_currency; Received_currency	Sending and receiving currencies
Sender_bank_location; Receiver_bank_location	Bank jurisdictions
Payment_type	Transaction channel or method
Is_laundering; Laundering_type	Binary label and transaction-pattern category

My unit of analysis will be the sender account-day which is one account's outgoing activity on one day, which matches what an investigator reviews. That gives 3,683,221 cases which 9,064 (about 1 in 406) involve laundering.

The dataset has two limits I'll account for. First, it's synthetic, so some patterns may be cleaner than they would be in real data. More importantly, it labels every laundering case, which a real bank wouldn't typically know. A bank only learns the outcome of cases it actually investigates and to keep my analysis realistic, my models will only learn from cases the bank would have reviewed, and I'll use the full labels only to evaluate results.

Hypotheses

Every hypothesis is tested under the same set of constraints 50 reviews per day that are scored daily from January through August 2023. The measure is how many laundering cases are caught within that limit, not accuracy. As we have talked about in class with other examples, laundering is so rare that a system flagging nothing would still be 99.75% accurate.

Hypothesis 1 (H1): A machine learning model catches more laundering than a rules-based system. I will compare calibrated rules against a logistic regression model and aLightGBM model. The models learn only from cases the bank reviewed in October - December 2022.

Hypothesis 2 (H2): Network-based features improve detection. I will add features about who an account pays and receives from which should help catch schemes that involve groups of accounts, like fan-ins and cycles and improve detection.

Hypothesis 3 (H3): A model that has a feedback loop and retrains monthly on new investigation results outperforms the same model left unchanged. At the end of each month, the bank will learn the outcomes of the cases it reviewed, and for my adaptive model it will learn from those outcomes and retrain each month.

Hypothesis 3 (H4) (future pilot): An AI investigation layer improves on the model alone at an acceptable cost per added case. Designed, not run.

For H1–H3, to check whether the differences are consistent rather than luck I will use bootstrap confidence intervals over days.

Project Scope and Methods

I will treat October to December 2022 as Kestrel's history and January to August 2023 as the period after it deploys the new systems. The work will run in five steps:

1. Audit and clean the data, and then build daily account features from Kestrel's history only, so no future information leaks in.
2. Build a rules benchmark from standard AML red flags like cash, cross-border payments, high-risk countries, unusual amounts which will be calibrated on Kestrel's history to make it a strong baseline.
3. Simulate the bank's past investigations which will be each days top 45 rule alerts plus 5 random cases from just below the cutoff (46- 200) and only these reviewed cases become training labels.
4. Compare and upgrade and test the rules against the different machine learning models (H1), then add network features (H2) and then add the monthly retraining feedback loop (H3) to the winner from (H1).
5. Translate results into business terms identifying cases caught, reviews and analyst hours saved under stated assumptions, and the added cost of governing a more complex system.

The project is split into five Python notebooks on GitHub 01 Audit, 02 cleaning, 03 EDA and features, 04 Modeling, and 05 feedback loop and network features, and a Tableau workbook with 3 dashboards, and a 10–12 slide deck for a 10-minute presentation to Kestrel's compliance leadership.

References

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